$\sum \frac{n}{(n+1)(n+2)(n+3)}$

$\alpha \sum \frac{1}{n+2} + \beta \sum \frac{1}{n+2}$

$\sum_{n=0}^{\infty} a_n + \sum_{n=0}^{\infty} a_{n+k}$

$= \sum_{n=0}^{k-1} a_n + 2 \sum_{n=k}^{+\infty} a_n$

$\sum_{n=1}^{\infty} \log\left(\frac{n^2+2n+1}{n^2}\right) = \sum_{1}^{\infty} \log\left(\frac{(n+2)^2}{n^2}\right) = \sum_{n=1}^{\infty} \log\left(\left(\frac{n+1}{n}\right)^2\right) = 2 \sum_{n=1}^{\infty} \log\left(\frac{n+1}{n}\right)$

$= 2 \sum_{1}^{N} [\log(n+1) - \log(n)] = 2 \left[\log(2) - \log(1) + \log(3) - \log(2) + \log(4) - \log(3) \dots + \log(N+1) - \log(N) \right]$

$2(\log N - \log(1))$

$N \to \infty \to +\infty$

$\sum_{n=1}^{\infty} \log\left(\frac{n^2+2n+1}{n^2}\right) = +\infty$

$\log\left(\frac{n^2+2n+1}{n^2}\right) = \log\left(1 + \frac{2}{n} + \frac{1}{n^2}\right) \to \log 1 = 0$

$\sum_{n=1}^{\infty} \frac{n \operatorname{arctg}(n+2) - (n+2)\operatorname{arctg}(n)}{n(n+2)}$

$\sum_{2}^{\infty} \frac{(n+1)\operatorname{arctg}(n+1)}{n(n+1)} - \sum \frac{(n+1)\operatorname{arctg} n}{n(n+1)}$

$= \sum_{2}^{\infty} \frac{\operatorname{arctg} n}{n} - \sum_{1}^{\infty} \frac{\operatorname{arctg} n}{n}$

$= -\frac{\operatorname{arctg}(1)}{1} = -\frac{\pi}{4}$

$\tan(\pi/4) = 1$

$\operatorname{arctg}(1) = \pi/4$

$1 = \frac{\sin t}{\cos t}$

